

AN ELEMENTARY RESOLUTION OF AN OEIS GENERATING-FUNCTION CONJECTURE

A marked-cycle proof of the trapezohedral domination conjecture

Classification, refined enumeration, log-concavity,
and asymptotic laws for OEIS A381190

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Abstract

Let a_n count vertex subsets of the n -trapezohedral graph that are minimal among all dominating sets and induce a connected subgraph. The OEIS entry A381190 records a rational generating function conjectured by Joerg Arndt. We prove this formula by classifying every counted set: it contains exactly one pole, exactly one vertex of the opposite ring class, and a cyclic binary word whose zero-to-zero gaps have lengths two or three. The opposite-class vertex marks an eligible edge of that word. Cutting at the marked gap gives

$$a_n = 2n(2p_{n-2} + p_{n-3}), \quad \sum_{m \geq 0} p_m x^m = \frac{1}{1 - x^2 - x^3}, \quad n \geq 3.$$

The same bijection gives an explicit size distribution, a trivariate generating function, strict log-concavity, a distinction between tree and unicyclic induced subgraphs, exact extremal counts, and an unbiased sampler. We also obtain a Binet formula, exponentially small absolute asymptotic error, two exact nearest-integer formulas, a closed finite binomial sum, and a central limit theorem for set size. Independent exhaustion of all 22,369,536 vertex subsets for $3 \leq n \leq 11$ agrees with the classification at the level of individual sets, not merely totals; the cases $3 \leq n \leq 10$ are independently reproduced by a second, separately written exhaustion, and is reproduced by a second, separately written exhaustive program in another language.

Status and scope. On 19 September 2026 the live OEIS entry still labels the generating function as conjectured, with attribution to Joerg Arndt on 7 January 2026 [1]. This manuscript supplies a complete self-contained proof of that assertion. Targeted searches did not locate an earlier proof; that is not a guarantee of literature priority. The proofs have been computationally cross-checked, but not formally verified in a proof assistant or independently peer-reviewed. The OEIS status was checked, but an exhaustive priority search was not performed; no claim of being the first proof anywhere in the literature is made. The computational material is independently reproducible, but is not a formal proof-assistant verification or a substitute for peer review. No OEIS edit or submission has been made.

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1 The problem and the answer

1.1 The graph and the meaning of minimality

For $n \geq 3$, write T_n for the graph with vertices

$$V(T_n) = \{N, S\} \cup \{a_i, b_i : i \in \mathbb{Z}/n\mathbb{Z}\}$$

and edges

$$Na_i, \quad Sb_i, \quad a_ib_i, \quad b_ia_{i+1} \quad (i \in \mathbb{Z}/n\mathbb{Z}). \quad (1.1)$$

Thus $a_0b_0a_1b_1 \cdots a_{n-1}b_{n-1}a_0$ is a ring of length $2n$; the pole N is joined to every a_i and the pole S to every b_i . This is the usual n -trapezohedral graph, with $2n + 2$ vertices and $4n$ edges; T_3 is the cube graph [2]. All indices below are cyclic. Sets are counted as subsets of this fixed graph, *not* up to rotation, reflection, or graph isomorphism.

Definition 1.1. A subset $D \subseteq V(G)$ is *dominating* if every vertex belongs to D or has a neighbor in D . It is *minimal dominating* if no proper subset of D is dominating. We call it a *connected minimal dominating set* when, in addition, the induced graph $G[D]$ is connected. Let $\mathcal{D}_n$ denote this family for T_n , and let $a_n = |\mathcal{D}_n|$.

The order of these qualifications matters. A *minimal connected dominating set* is minimal only within the family of connected dominating sets. Such a set can cease to be connected when one vertex is deleted while remaining dominating, so it need not be minimal dominating. This distinction is explicit in the standard definitions [3, 4]. Here every selected vertex must be indispensable *for domination itself*.

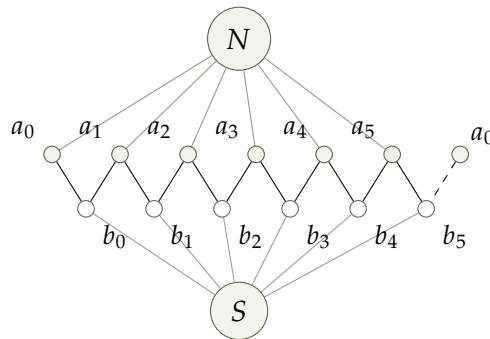


Figure 1: An unrolled drawing of T_6 . The rightmost a_0 repeats the leftmost vertex; it is not an additional vertex. The repeated boundary makes the ring and the two alternating pole neighborhoods explicit.

1.2 The conjecture and a simpler formula

The entry A381190 was contributed by Eric W. Weisstein on 16 February 2025. The retrieved entry lists 40 terms, beginning

$$6, 16, 30, 36, 70, 96, 144, 220, 308, 456, 650, 924, \dots$$

with offset $n = 3$, and gives Arndt's conjecture [1]

$$\sum_{n \geq 3} a_n x^n = \frac{-2x^3(4x^5 + 8x^4 + 4x^3 - 9x^2 - 8x - 3)}{(x^3 + x^2 - 1)^2}. \quad (1.2)$$

We will prove the following stronger and more transparent statement.

Theorem 1.2 (Main enumeration). *Let p_m count ordered compositions of m into parts 2 and 3, including the empty composition when $m = 0$. Put $p_m = 0$ for $m < 0$. Then*

$$a_n = 2n(2p_{n-2} + p_{n-3}) \quad (n \geq 3). \quad (1.3)$$

Equivalently, if $P(m)$ is A000931 with its OEIS indexing, then

$$a_n = 2n(2P(n+1) + P(n)). \quad (1.4)$$

The indexing in (1.4) uses $p_m = P(m+3)$; the composition interpretation and that shift are recorded in A000931 [5]. The sequence p_m is specified independently by

$$p_0 = 1, \quad p_1 = 0, \quad p_2 = 1, \quad p_m = p_{m-2} + p_{m-3} \quad (m \geq 3), \quad \sum_{m \geq 0} p_m x^m = \frac{1}{1 - x^2 - x^3}. \quad (1.5)$$

The recurrence follows by deleting the first part of a nonempty composition. Its initial terms are 1, 0, 1, 1, 1, 2, 2, 3, 4, 5, ...

2 A structural classification

2.1 Private neighbors

For a vertex w , let $N[w]$ denote its closed neighborhood: w together with all vertices adjacent to w .

Lemma 2.1 (Private-neighbor criterion). *A dominating set D is minimal if and only if every $v \in D$ has a vertex w with*

$$N[w] \cap D = \{v\}. \quad (2.1)$$

The vertex w may equal v and is called a private neighbor of v relative to D .

Proof. If (2.1) holds, deleting v leaves w undominated. Thus no one-vertex deletion preserves domination, and no proper subset can dominate. Conversely, if no such w exists for some v , every vertex dominated by v is also dominated by another vertex of D . Hence $D \setminus \{v\}$ still dominates, contradicting minimality. $\square$

2.2 Exactly one pole survives

Lemma 2.2. *Every $D \in \mathcal{D}_n$ contains exactly one of N and S .*

Proof. The pair $\{N, S\}$ dominates T_n , so a minimal dominating set containing both poles must equal that pair. But the poles are not adjacent, and the pair is disconnected.

Suppose instead that D contains neither pole. Its induced graph lies in the ring C_{2n} . A nonempty connected proper subset of a cycle is a consecutive path. If this path has ℓ vertices, its closed neighborhood within the ring has at most $\ell + 2$ vertices. Because there are no selected poles to dominate the remaining ring vertices, $\ell \geq 2n - 2 \geq 4$. In particular, the path contains at least two vertices of each alternating class.

Take an internal vertex v of the path. Each of its two ring neighbors belongs to D , so neither v nor either ring neighbor can be private for v . Its only other neighbor is its pole, which is adjacent to at least two selected vertices of the same class. The pole is not private either. Thus v has no private neighbor, contradicting theorem 2.1. If D is the whole ring, the same argument applies to every ring vertex. $\square$

By symmetry we may now assume $N \in D$ and $S \notin D$. The explicit symmetry $N \leftrightarrow S, a_i \mapsto b_{-i}, b_i \mapsto a_{-i}$ preserves all edges in (1.1).

Lemma 2.3. *If $D \in \mathcal{D}_n$ contains N , then it contains exactly one b -vertex.*

Proof. At least one b -vertex is needed to dominate the unselected pole S . Every selected b_i must have a selected a -neighbor, since S is absent and D is connected. Therefore b_i cannot be its own private neighbor. Neither adjacent a -vertex can be private for it, because both are adjacent to the selected pole N . Its only remaining possible private neighbor is S . If two or more b -vertices were selected, S would be private for none of them. Minimality therefore forces exactly one, say b_j . $\square$

2.3 A cyclic word with one marked edge

Write

$$D = \{N, b_j\} \cup \{a_i : x_i = 1\}, \quad x_i \in \{0, 1\}. \quad (2.2)$$

View $x_0x_1 \cdots x_{n-1}$ as a word on the auxiliary cycle with vertices $a_0, \dots, a_{n-1}$. Its edge $e_j = (a_j, a_{j+1})$ is distinguished by the selected vertex b_j .

Theorem 2.4 (Classification). *The set (2.2) belongs to $\mathcal{D}_n$ if and only if the following hold.*

- (i) *The cyclic word contains neither 00 nor 111.*
- (ii) *If the marked edge has endpoints 11, it is the middle edge of a cyclic 0110 pattern.*
- (iii) *If the marked edge has endpoints 01 or 10, its endpoint labeled 1 is the middle vertex of a cyclic 010 pattern.*

In (ii), the surrounding zeros already follow from (i). A cyclic pattern can wrap around the end of the word; for $n = 3$ the two displayed zeros of 0110 are the same vertex.

Proof. Every unselected b_i needs a selected a -neighbor in order to be dominated. The selected b_j also needs a selected a -neighbor, now for connectivity. Consequently $x_i + x_{i+1} \geq 1$ for every i , excluding 00.

Consider a selected a_i . Its closed neighborhood consists of a_i, N, b_{i-1} , and b_i . Neither a_i nor N can be private for a_i , because N is selected. The distinguished b_j is not private for a_i , because it is

itself selected. An unselected adjacent b -vertex is private for a_i exactly when its other a -neighbor is unselected. We obtain the exact rule:

Every 1 must have a neighboring 0 across an edge other than the marked edge.

This excludes 111. If the marker is 11, each of the two marked endpoints must find its zero on its other side, giving 0110. If the marker is 10 or 01, its 1 must find a second zero on the other side, giving 010. All other ones merely need one adjacent zero, which the absence of 111 ensures. This proves necessity.

Conversely, suppose the stated word conditions hold. Every a -vertex is dominated by N , every b -vertex has a selected a -neighbor by the absence of 00, and S is dominated by b_j . All selected a -vertices are adjacent to N , while b_j is adjacent to at least one of them, so D is connected.

Every selected a_i has a private neighboring b -vertex by the exact rule above. The vertex S is private for b_j . It remains to supply a private neighbor for N . If the marker is 11, any zero of the word is an unselected a -vertex not incident to the marker; its closed neighborhood meets D only in N . At least one zero exists because 111 is forbidden and $n \geq 3$. If the marker is 01 or 10, use the other zero in the required 010 pattern. The two neighbors of its middle vertex are distinct for $n \geq 3$, and that other zero is not incident to the marked edge. Again it is private for N . Thus every selected vertex has a private neighbor, proving minimality. $\square$

Example 2.5 (A complete certificate at $n = 5$). Let $n = 5$ and take the cyclic word $x_0x_1x_2x_3x_4 = 01101$ with the marked edge $e_1 = (a_1, a_2)$, so that the selected opposite-class vertex is b_1 and

$$D = \{N, a_1, a_2, a_4, b_1\}.$$

The word has no 00 and no 111, and the marker has endpoints 11 sitting as the middle edge of the cyclic pattern $x_0x_1x_2x_3 = 0110$; conditions (i) and (ii) of [theorem 2.4](#) therefore hold. A complete private-neighbor certificate is

Selected vertex	N	a_1	a_2	a_4	b_1
Private neighbor	a_0	b_0	b_2	b_3	S

Each column is checked directly from (1.1). For instance $N[a_0] = \{a_0, N, b_4, b_0\}$ meets D only in N , and $N[b_0] = \{b_0, S, a_0, a_1\}$ meets D only in a_1 ; the pole S is adjacent to b_1 alone among the selected vertices. The zero set is $Y = \{0, 3\}$, so $k = 2$ with one gap of each length, and (4.4) predicts $4k - n = 3$ eligible markers. Counting the five labeled rotations, the three markers, and the two poles gives $a_5 = 2 \cdot 5 \cdot 3 = 30$, in agreement with (4.5) below.

The archived file `data/certificates_n5.json` lists all thirty members of $\mathcal{D}_5$ together with such a witness for every selected vertex. That file and the standard-library verifier that produced it use the one-based labels $u, v, a_1, \dots, a_n, b_1, \dots, b_n$ of the incoming package ([section C](#)); the dictionary to the labels used here is $u = N, v = S, a_i = a_{i-1}, b_i = b_{i-1}$. Under it, the certificate above is the row $D = \{u, a_2, a_3, a_5, b_2\}$ of that file.

2.4 The gap formulation

Let $Y = \{i : x_i = 0\}$. The absence of 00 means that Y is independent in C_n ; the absence of 111 means that every vertex outside Y is adjacent to Y . Thus Y is a *maximal independent set* of C_n . In the cyclic

order, distances between consecutive zeros are exactly 2 or 3. These are the two possible blocks 01 and 011, with the next block starting at the next zero.

Corollary 2.6 (Marked gaps). *In a zero-to-zero gap of length 2, both edges are eligible markers. In a gap of length 3, only the middle edge is eligible. There are no other possibilities.*



Figure 2: Eligible marked edges are thick. Adjacent gaps share their endpoint zero. When there is only one gap, its two displayed endpoint zeros coincide on the cycle.

The unmarked objects are classical: maximal independent sets of C_n are counted by the Perrin numbers [6]. The essential step here is the precise edge-marking condition imposed by private neighbors in T_n .

3 Cutting at the marker: proof of the conjecture

3.1 The rooted composition bijection

Fix the selected pole N and the selected opposite-class vertex b_0 . There are three possible local marker configurations. If the marked edge is 11, it sits inside one length-3 gap. Remove that anchored gap and read the remaining gaps in the fixed cyclic orientation. They form an ordered composition of $n - 3$ into 2's and 3's, and the operation is reversible. Hence there are p_{n-3} such words.

If the marker has type 01, it is one of the two edges of a length-2 gap. Removing that anchored gap leaves a composition of $n - 2$. The same is true for type 10, but its orientation relative to the fixed marker is different. These cases are disjoint and contribute $2p_{n-2}$.

This argument includes the empty remainder. In particular, for $n = 3$ the length-3 gap can occupy the whole cycle, contributing $p_0 = 1$ for a fixed pole and marker. The length-2 alternatives contribute $p_1 = 0$.

There are two choices of the selected pole and n choices of its unique opposite-class vertex. These choices are recoverable from D , so they cannot create overcounting. This proves [theorem 1.2](#).

The same bijection can be read blockwise instead of gapwise: starting at the initial zero of the marked gap and reading in the fixed cyclic orientation produces a *marked first block* — a block 01 carrying either of its two marks, or a block 011 carrying its unique mark, which is exactly the numerator $2x^2 + x^3$ of (3.1) below — followed by a free ordered list of unmarked blocks, which is exactly the factor $1/(1 - x^2 - x^3)$. In either reading the mark singles out one specific block, so a periodic word creates no rotational ambiguity and no division by an orbit size is needed.

3.2 The generating function

Introduce

$$R(x) = \frac{2x^2 + x^3}{1 - x^2 - x^3}. \quad (3.1)$$

For $n \geq 3$, its x^n coefficient is $a_n/(2n)$. There is also an artificial x^2 coefficient equal to 2; no graph with $n = 2$ is included in this problem. Since $x \frac{d}{dx}$ multiplies the x^n coefficient by n ,

$$\begin{aligned} A(x) &:= \sum_{n \geq 3} a_n x^n = 2xR'(x) - 8x^2 \\ &= \frac{-2x^3(4x^5 + 8x^4 + 4x^3 - 9x^2 - 8x - 3)}{(x^3 + x^2 - 1)^2}. \end{aligned} \quad (3.2)$$

This is exactly (1.2). The squared denominator is explained structurally: after fixing a pole, an eligible location is marked on a cyclic object, introducing a factor of n into a Padovan-type count.

n	p_{n-2}	p_{n-3}	$a_n/(2n)$	a_n
3	0	1	1	6
4	1	0	2	16
5	1	1	3	30
6	1	1	3	36
7	2	1	5	70
8	2	2	6	96
9	3	2	8	144
10	4	3	11	220
11	5	4	14	308
12	7	5	19	456
13	9	7	25	650
14	12	9	33	924

Table 1: Initial values computed from the bijection. These agree with the corresponding published A381190 terms [1].

4 Exact size and topology refinements

4.1 Counting fixed gap multiplicities

Let $k = |Y|$ be the number of zeros, let s be the number of length-2 gaps, and let t be the number of length-3 gaps. Then

$$s + t = k, \quad 2s + 3t = n, \quad s = 3k - n, \quad t = n - 2k. \quad (4.1)$$

Consequently

$$\left\lceil \frac{n}{3} \right\rceil \leq k \leq \left\lfloor \frac{n}{2} \right\rfloor, \quad |D| = n - k + 2. \quad (4.2)$$

The number of labeled cyclic zero sets with these multiplicities is

$$\frac{n}{k} \binom{k}{t}. \quad (4.3)$$

Indeed, distinguish a zero and select its index in n ways; choose the positions of the t long gaps among the k ordered gaps in $\binom{k}{t}$ ways. This counts each unrooted zero set exactly k times, once per

choice of its distinguished zero. The reasoning is valid even when the gap word is periodic: the root is an actual labeled cycle vertex, not a necklace orbit.

By [theorem 2.6](#), a zero set has

$$2s + t = 4k - n \quad (4.4)$$

eligible markers. This number depends only on n and k .

Theorem 4.1 (The complete size distribution). *For k in (4.2), the number $c_{n,k}$ of sets in $\mathcal{D}_n$ of cardinality $n - k + 2$ is*

$$\begin{aligned} c_{n,k} &= \frac{2n(4k - n)}{k} \binom{k}{n - 2k} \\ &= 2n \left(2 \binom{k - 1}{n - 2k} + \binom{k - 1}{n - 2k - 1} \right). \end{aligned} \quad (4.5)$$

Outside that range, the count is zero. A binomial coefficient is interpreted as zero when its lower index is negative or exceeds its nonnegative upper index.

Proof. Multiply (4.3) by the eligible-marker count (4.4) and by the two choices of pole. For the second form, use $(s/k) \binom{k}{t} = \binom{k-1}{t} \binom{k-1}{t-1}$ and $(t/k) \binom{k}{t} = \binom{k-1}{t-1}$, with $2s + t = 4k - n$. $\square$

Summing [theorem 4.1](#) over the admissible sizes turns the Padovan recurrence of [theorem 1.2](#) into a closed finite sum, with no recurrence and no generating function.

Corollary 4.2 (A finite binomial sum for the total). *For every $n \geq 3$, with s and t the numbers of length-2 and length-3 gaps as in (4.1),*

$$a_n = 2n \sum_{\substack{s,t \geq 0 \\ 2s+3t=n}} \left(2 \binom{s+t-1}{t} + \binom{s+t-1}{t-1} \right) = 2n \sum_{\substack{s,t \geq 0 \\ 2s+3t=n}} \frac{2s+t}{s+t} \binom{s+t}{t}. \quad (4.6)$$

Proof. Each pair (s, t) with $2s + 3t = n$ corresponds to the single admissible $k = s + t$ of (4.1), and the summand is $c_{n,k}/(2n)$ in the two forms of (4.5). Summing over k gives a_n . The two summands agree term by term because, writing $k = s + t$ and $s = k - t$,

$$2 \binom{k-1}{t} + \binom{k-1}{t-1} = \frac{2(k-t) + t}{k} \binom{k}{t} = \frac{2s+t}{k} \binom{k}{t}. \quad (4.7)$$

$\square$

Although the second form of (4.6) displays a quotient, every summand of the first form is visibly a nonnegative integer. This gives a direct combinatorial reason, independent of (5.5) below, that $a_n/(2n)$ is an integer: it is the number of completions once one pole and one marker block have been fixed.

4.2 Only two induced graph shapes occur

In (2.2), the pole N is joined to all $n - k$ selected a -vertices. The vertex b_j is joined to either one or two of them, depending on its marker. There are no other edges in the induced graph.

Corollary 4.3 (Trees and four-cycles). *Every member of $\mathcal{D}_n$ induces either a tree or a unicyclic graph whose unique cycle has length 4. For $t = n - 2k$, their respective counts at size $n - k + 2$ are*

$$T_{n,k} = 4n \binom{k-1}{t}, \quad U_{n,k} = 2n \binom{k-1}{t-1}. \quad (4.8)$$

In total,

$$T_n = 4np_{n-2}, \quad U_n = 2np_{n-3}, \quad a_n = T_n + U_n. \quad (4.9)$$

Proof. A marker 01 or 10 attaches b_j as a leaf to one leaf of the pole-centered star. A marker 11 adds the path $a_j b_j a_{j+1}$ between two leaves and hence creates precisely the cycle $Na_j b_j a_{j+1} N$. A fixed zero set has $2s$ tree markers and t unicyclic markers. Multiplying as before proves (4.8); the rooted-composition bijection proves (4.9) directly. $\square$

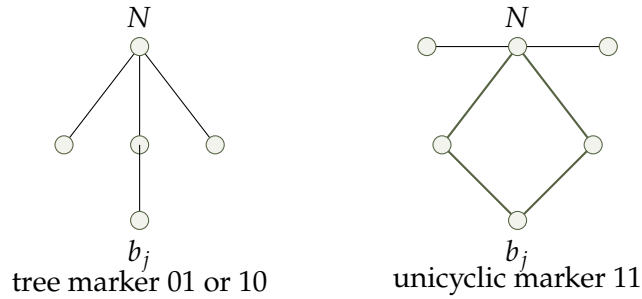


Figure 3: Representative induced shapes. More selected a -vertices simply add leaves adjacent to the pole; no further cycles can arise.

4.3 A trivariate generating function

Let y mark the size of D , and let v mark the unicyclic type. Define

$$f_n(y, v) = \sum_{D \in \mathcal{D}_n} y^{|D|} v^{\mathbf{1}\{T_n[D] \text{ unicyclic}\}}, \quad F(x, y, v) = \sum_{n \geq 3} f_n(y, v) x^n. \quad (4.10)$$

A length-2 gap contains one selected a -vertex and has weight yx^2 . A length-3 gap contains two and has weight y^2x^3 . The chosen pole and opposite-class vertex contribute another factor y^2 . The marker belongs either to a length-2 gap, with two orientations, or to a length-3 gap, with one unicyclic orientation. Thus

Theorem 4.4 (Joint generating function).

$$F(x, y, v) = 2y^2x \frac{\partial}{\partial x} \left(\frac{2yx^2 + vy^2x^3}{1 - yx^2 - y^2x^3} \right) - 8y^3x^2. \quad (4.11)$$

The derivative accounts for the n possible markers, the outer 2 for the pole, and the last term removes the artificial $n = 2$ contribution. Writing $f_n(y) = f_n(y, 1)$, the first examples are

$$\begin{aligned} f_3(y) &= 6y^4, & f_4(y) &= 16y^4, & f_5(y) &= 30y^5, \\ f_6(y) &= 24y^5 + 12y^6, & f_7(y) &= 70y^6, & f_8(y) &= 32y^6 + 64y^7. \end{aligned}$$

The unexpanded derivative form of (4.11) retains the combinatorial meaning more clearly than its expanded rational numerator. For symbolic use, however, the size-refined specialization does have a compact closed form.

Corollary 4.5 (Expanded size-refined generating function). *Carrying out the differentiation in (4.11) at $v = 1$ gives*

$$F(x, y, 1) = \sum_{n \geq 3} f_n(y) x^n = \frac{-2x^3 y^4 (4x^5 y^3 + 8x^4 y^2 + 4x^3 y - 9x^2 y - 8x - 3)}{(x^3 y^2 + x^2 y - 1)^2}. \quad (4.12)$$

Setting $y = 1$ recovers (1.2).

The substitution $x \mapsto x$, $y \mapsto 1$ sends the denominator $(x^3 y^2 + x^2 y - 1)^2$ to $(x^3 + x^2 - 1)^2$ and the numerator to that of (1.2), so (4.12) is a genuine refinement of the conjectured formula rather than a separate claim. It is the convenient form for automatic coefficient extraction; (4.11) is the convenient form for reading off the combinatorics.

4.4 A weighted Perrin identity

Let

$$I_n(z) = \sum_{Y \text{ maximal independent in } C_n} z^{|Y|}.$$

From the rooted-gap count (4.3),

$$\begin{aligned} H(x, z) &:= \sum_{n \geq 3} I_n(z) x^n = x \frac{\partial}{\partial x} \sum_{k \geq 1} \frac{(zx^2 + zx^3)^k}{k} - 2zx^2 \\ &= \frac{2zx^2 + 3zx^3}{1 - zx^2 - zx^3} - 2zx^2. \end{aligned} \quad (4.13)$$

Because a zero set with k vertices contributes $2(4k - n)$ marked objects,

$$a_n = 2(4I'_n(1) - nI_n(1)), \quad A(x) = (8z\partial_z - 2x\partial_x)H(x, z)|_{z=1}. \quad (4.14)$$

At $z = 1$, $I_n(1)$ is the usual Perrin cycle count [6]. The new enumeration is therefore a specific weighted first moment of that classical family, not an unrelated recurrence coincidence.

5 Recurrences and exact extremal information

5.1 A third-order recurrence after normalization

Let $b_n = a_n/(2n)$ for $n \geq 3$. By (1.3),

$$b_3 = 1, \quad b_4 = 2, \quad b_5 = 3, \quad b_n = b_{n-2} + b_{n-3} \quad (n \geq 6). \quad (5.1)$$

Equivalently,

$$(n-2)(n-3)a_n = n(n-3)a_{n-2} + n(n-2)a_{n-3} \quad (n \geq 6). \quad (5.2)$$

The size polynomials satisfy the parallel recurrence

$$\frac{f_n(y)}{n} = y \frac{f_{n-2}(y)}{n-2} + y^2 \frac{f_{n-3}(y)}{n-3} \quad (n \geq 6). \quad (5.3)$$

The same equation holds with $f_n(y, v)$ in place of $f_n(y)$. These formulas follow by multiplying the normalized generating function by $1 - yx^2 - y^2x^3$, or by deleting the first unmarked gap.

Without normalization, the constant-coefficient recurrence is

$$a_n = 2a_{n-2} + 2a_{n-3} - a_{n-4} - 2a_{n-5} - a_{n-6} \quad (n \geq 9). \quad (5.4)$$

The starting index 9 is essential: the numerator in (1.2) has degree 8. The six initial values are $a_3 = 6, a_4 = 16, a_5 = 30, a_6 = 36, a_7 = 70, a_8 = 96$.

In fact, order 6 is the minimal order of an eventual constant-coefficient homogeneous recurrence. Write $Q(x) = 1 - x^2 - x^3$ and $\Pi(x) = 2x^2 + x^3 = x^2(2 + x)$, so that $R = \Pi/Q$ in (3.1). Each step is a concrete evaluation.

- The roots of Q are simple. Indeed $Q'(x) = -x(2 + 3x)$ vanishes only at $x = 0$ and $x = -2/3$, while $Q(0) = 1$ and $Q(-2/3) = 23/27$ are both nonzero, so Q and Q' have no common root.
- Π and Q have no common root. The roots of Π are 0 and -2 , while $Q(0) = 1$ and $Q(-2) = 5$.
- At a root x_0 of Q the numerator $2x(\Pi'Q - \Pi Q')$ of $2xR'$ takes the value $-2x_0\Pi(x_0)Q'(x_0)$, which is nonzero by the two preceding points and $x_0 \neq 0$. Hence $2xR'$ has a genuine double pole at each of the three roots, and subtracting the polynomial $8x^2$ cannot remove it.

Thus the reduced denominator of A is Q^2 , of degree 6. An eventual constant-coefficient recurrence with characteristic polynomial Φ makes $\Phi(x)A(x)$ a polynomial, so Q^2 divides Φ and $\deg \Phi \geq 6$; no lower-order recurrence can suffice. This is a statement about *constant* coefficients only, and does not conflict with the three-term polynomial-coefficient recurrence (5.2).

The factor $2n$ in (4.5) proves the stronger divisibility

$$2n \mid c_{n,k}, \quad 2n \mid T_{n,k}, \quad 2n \mid U_{n,k}. \quad (5.5)$$

Fixing the pole and marker also explains this divisibility combinatorially: all $2n$ such choices have the same number of completions at each size and type.

5.2 Smallest and largest possible sets

Theorem 5.1 (Extremal sizes and their multiplicities). *Every integer size in the interval*

$$\left\lceil \frac{n}{2} \right\rceil + 2 \leq |D| \leq \left\lfloor \frac{2n}{3} \right\rfloor + 2 \quad (5.6)$$

occurs in $\mathcal{D}_n$. The number of sets attaining the lower endpoint is

$$\begin{cases} 4n, & n \text{ even}, \\ 2n(n-2), & n \text{ odd}. \end{cases} \quad (5.7)$$

For the upper endpoint, writing $n = 3m + r$ with $r \in \{0, 1, 2\}$, the count is

$$\begin{cases} 2n, & r = 0, \\ nm(m+3), & r = 1, \\ 2n(m+2), & r = 2. \end{cases} \quad (5.8)$$

Proof. The admissible k in (4.2) are consecutive and each has $s, t \geq 0$, so a corresponding gap word exists. Its marker count $2s + t$ is positive. Since $|D| = n - k + 2$, this proves (5.6) and attainment. For the smallest size use $k = \lfloor n/2 \rfloor$: $t = 0$ for even n , and $t = 1$ for odd n . Substitution in (4.5) gives (5.7). For the largest size use $k = \lceil n/3 \rceil$. When $n = 3m, 3m + 1, 3m + 2$, the pairs (s, t) are respectively $(0, m), (2, m - 1), (1, m)$. The same formula simplifies to (5.8). $\square$

5.3 A growing gap between two notions of minimality

Let $\gamma_c(T_n)$ be the ordinary connected domination number: the minimum size of any connected dominating set, without requiring minimal domination.

Proposition 5.2. *For every $n \geq 3$, $\gamma_c(T_n) = 4$, whereas the smallest connected minimal dominating set has size $\lceil n/2 \rceil + 2$.*

Proof. The set $\{N, a_0, b_0, S\}$ is connected and dominating, proving the upper bound 4. A connected dominating set containing both poles needs at least two ring vertices to connect them, since their distance is 3. If it contains neither pole and has at most three vertices, it is a ring path and dominates at most five ring vertices, fewer than $2n$. If it contains exactly one pole, say N , and has at most three vertices, it needs a selected b -vertex to dominate S and a selected a -vertex to connect that b to N . Thus it consists of N and one adjacent a, b pair. The selected a dominates only its two adjacent b -vertices, one of which is the selected b itself; a third b -vertex is undominated. Hence no such set exists. The second assertion is theorem 5.1. $\square$

Thus confusing the two families would not produce a harmless constant-factor error: their smallest cardinalities separate linearly with n .

6 Strict log-concavity of the size distribution

Theorem 6.1. *For fixed n , the positive sequence $(c_{n,k})$ on the interval (4.2) is strictly log-concave at every interior index:*

$$c_{n,k}^2 > c_{n,k-1}c_{n,k+1}. \quad (6.1)$$

The same is true for the nonzero coefficients of $f_n(y)$, read in increasing size. In particular the distribution is unimodal, with either one mode or two adjacent equal modes. If the support has at most two points, the strict log-concavity assertion has no interior inequality to check.

Proof. For two consecutive admissible indices, direct cancellation in (4.5) gives

$$\begin{aligned} \frac{c_{n,k+1}}{c_{n,k}} &= \frac{4k+4-n}{4k-n} \frac{k(n-2k)(n-2k-1)}{(3k-n+1)(3k-n+2)(3k-n+3)} \\ &= \left(1 + \frac{4}{4k-n}\right) \frac{k}{3k-n+3} \frac{n-2k}{3k-n+1} \frac{n-2k-1}{3k-n+2}. \end{aligned} \quad (6.2)$$

Every displayed factor is positive on the transition interval. The first is strictly decreasing. The second has derivative $(3 - n)/(3k - n + 3)^2 \leq 0$. The third and fourth are strictly decreasing because their numerators decrease and their positive denominators increase. Thus the ratio in (6.2) strictly decreases with k whenever two successive ratios are defined. This is exactly strict log-concavity. Replacing k by the reversed index $|D| = n - k + 2$ preserves the inequalities. Decreasing positive consecutive ratios can cross 1 only once, giving the stated description of the modes. $\square$

This result is an exact finite- n statement. It is stronger than the visual bell shape suggested by the asymptotic normal law proved below and does not rely on that approximation.

7 Binet formula, asymptotics, and exact rounding

7.1 Three characteristic roots

Let α be the positive real root of

$$\alpha^3 = \alpha + 1, \quad \alpha = 1.324717957244746025960908854478 \dots, \quad (7.1)$$

and let $\beta, \bar{\beta}$ be the other two roots. The polynomial $z^3 - z - 1$ has three distinct roots and exactly one real root: at its two real critical points the polynomial is negative, while it tends to opposite infinities at the two ends of the real line. By Vieta's formulas,

$$\beta + \bar{\beta} = -\alpha, \quad \beta\bar{\beta} = \alpha^{-1}, \quad |\beta| = \alpha^{-1/2} < 1. \quad (7.2)$$

Write

$$C(z) = \frac{2z + 1}{2z + 3}, \quad K = 2C(\alpha) = 1.291964709301166107055203050684 \dots \quad (7.3)$$

Theorem 7.1 (Exact Binet formula). *For every $n \geq 3$,*

$$a_n = 2n \left(C(\alpha)\alpha^n + C(\beta)\beta^n + C(\bar{\beta})\bar{\beta}^n \right). \quad (7.4)$$

Consequently

$$a_n = Kn\alpha^n + O(n\alpha^{-n/2}), \quad (7.5)$$

and, more explicitly,

$$|a_n - Kn\alpha^n| \leq 4n|C(\beta)|\alpha^{-n/2}, \quad |C(\beta)| = 0.580110540477999273668313348466 \dots \quad (7.6)$$

Proof. At a pole $x = 1/\lambda$ of $R(x)$, where $\lambda^3 = \lambda + 1$, the coefficient of $(1 - \lambda x)^{-1}$ in its partial fraction decomposition is

$$\frac{2 + 1/\lambda}{2 + 3/\lambda} = C(\lambda).$$

Because numerator and denominator of R have equal degree, there is also the constant polynomial part -1 . Hence

$$R(x) = -1 + \sum_{\lambda \in \{\alpha, \beta, \bar{\beta}\}} \frac{C(\lambda)}{1 - \lambda x}. \quad (7.7)$$

For positive coefficient indices the constant is irrelevant. Extracting the x^n coefficient and using (1.3) proves (7.4). The conjugate pair and (7.2) give (7.6) by the triangle inequality, and then (7.5) follows. $\square$

The error here tends to zero *absolutely*, despite the exponential growth of a_n . Relative to the main term it is $O(\alpha^{-3n/2})$. There is no hidden series of inverse powers of n : the entire dominant contribution is exactly $Kn\alpha^n$, and the remainder consists of the two explicit, exponentially decaying oscillatory terms in (7.4).

7.2 An exact nearest-integer formula

Theorem 7.2. *For all $n \geq 4$,*

$$a_n = 2n \left\lfloor C(\alpha)\alpha^n + \frac{1}{2} \right\rfloor. \quad (7.8)$$

The exceptional value is $a_3 = 6$; the same rounding expression at $n = 3$ would give 12.

Proof. For $b_n = a_n/(2n)$, (7.4) gives

$$|b_n - C(\alpha)\alpha^n| \leq 2|C(\beta)|\alpha^{-n/2}.$$

The bound decreases with n , and it is already less than $1/2$ at $n = 6$. Here is an exact certificate, rather than a numerical assumption. From (7.2),

$$|C(\beta)|^2 = \frac{4 - 2\alpha^2 + \alpha}{4 - 6\alpha^2 + 9\alpha}.$$

The desired inequality at $n = 6$ is $16|C(\beta)|^2 < \alpha^6$. Reducing its numerator with $\alpha^3 = \alpha + 1$ leaves

$$42\alpha^2 - 8\alpha - 63 > 0.$$

The rational interval $13247/10000 < \alpha < 13248/10000$ is certified by the signs of $z^3 - z - 1$ at its endpoints and monotonicity for $z > 1$. The last quadratic is increasing on this interval, and at its lower endpoint it is exactly $5263189/50000000 > 0$. The denominator above is positive on the same interval. This proves the bound for every $n \geq 6$.

The remaining cases follow by rational interval substitution in the increasing function $C(z)z^n$ for $z > 1$:

$$1.98 < C(\alpha)\alpha^4 < 2.00, \quad 2.63 < C(\alpha)\alpha^5 < 2.64.$$

Their nearest integers are $b_4 = 2$ and $b_5 = 3$. Finally, $1.50 < C(\alpha)\alpha^3 < 1.51$, whose nearest integer is 2, whereas $b_3 = 1$. $\square$

Theorem 7.2 rounds the *normalized* count $b_n = a_n/(2n)$. One can also round a_n itself, at the cost of a much later threshold.

Corollary 7.3 (Rounding the unnormalized count). *For every integer $n \geq 45$,*

$$a_n = \left\lfloor Kn\alpha^n + \frac{1}{2} \right\rfloor, \quad K = 2C(\alpha). \quad (7.9)$$

The threshold 45 is sufficient; no minimality claim is made for it.

Proof. First, $|C(\beta)| < 1$, by an exact comparison rather than by its decimal value. With the expression for $|C(\beta)|^2$ displayed in the proof of theorem 7.2, the denominator $4 - 6\alpha^2 + 9\alpha$ exceeds the numerator $4 - 2\alpha^2 + \alpha$ by $4\alpha(2 - \alpha) > 0$, and the denominator is positive on the certified interval $13247/10000 < \alpha < 13248/10000$. Hence (7.6) weakens to

$$|a_n - Kn\alpha^n| < 4n\alpha^{-n/2}. \quad (7.10)$$

Next, $\alpha > 64/49$. The function $t^3 - t - 1$ is increasing for $t > 1$, and

$$\left(\frac{64}{49}\right)^3 - \frac{64}{49} - 1 = -\frac{9169}{117649} < 0,$$

so the real root lies above $64/49$. Therefore $\alpha^{-1/2} < 7/8$ and (7.10) gives $|a_n - Kn\alpha^n| < 4n(7/8)^n$. The right-hand side decreases for integer $n \geq 8$, since $\frac{n+1}{n} \cdot \frac{7}{8} < 1$ exactly when $n > 7$. At $n = 45$ the inequality $4 \cdot 45 \cdot 7^{45} < \frac{1}{2} \cdot 8^{45}$ is the integer statement

$$360 \cdot 7^{45} < 8^{45}, \quad (7.11)$$

which holds. Hence the error is strictly below $1/2$ for every $n \geq 45$, and since a_n is an integer, (7.9) follows. $\square$

Remark 7.4. Theorem 7.2 and theorem 7.3 round *different* quantities, and neither threshold corrects the other. The normalized error $|b_n - C(\alpha)\alpha^n| \leq 2|C(\beta)|\alpha^{-n/2}$ carries no factor of n and falls below $1/2$ already at $n = 6$, with $n = 4, 5$ settled by rational interval substitution; only $n = 3$ is exceptional. The unnormalized error carries the factor n , so it decays only after n has been overtaken by $\alpha^{-n/2}$; the table below shows it is still about 2.70 at $n = 20$, and (7.9) does fail for many indices below its threshold, the largest such index being $n = 34$. A threshold well beyond the normalized one is therefore genuinely needed for the unnormalized form. Which statement is more useful depends on the purpose: (7.8) recovers essentially every term, while (7.9) expresses a_n itself by a single closed expression in one real algebraic number.

n	a_n	$a_n - Kn\alpha^n$ (approximately)
20	7160	2.701743849
45	18197730	-0.100362733
50	82488200	-0.083242473
100	210665294616400	0.000181710

Table 2: The absolute error in (7.5), illustrating theorem 7.4. These decimals are diagnostics; the uniform statements are theorem 7.2 and theorem 7.3.

7.3 The limiting induced topology

From (4.9) and the same dominant-root calculation,

$$\frac{U_n}{a_n} \longrightarrow \frac{1}{2\alpha + 1} = 0.274014950099455797655243575296 \dots, \quad \frac{T_n}{a_n} \longrightarrow \frac{2\alpha}{2\alpha + 1}. \quad (7.12)$$

Thus the limiting unicyclic proportion is approximately 27.4015%. This is a limit of exact labeled-set counts, not a simulation estimate.

8 The distribution of a random set's size

Choose D uniformly from $\mathcal{D}_n$, and let $L_n = |D|$. The precise size counts allow exact moments at every n . They also give a simple asymptotic law.

Theorem 8.1 (Size central limit theorem). *Define*

$$\begin{aligned}\mu &= \frac{\alpha + 2}{2\alpha + 3} = 0.588504411337354236618099618664 \dots, \\ \sigma^2 &= \frac{\alpha^4}{(2\alpha + 3)^3} = 0.017079629057536730116313074817 \dots, \\ \delta &= 2 - \frac{4\alpha^4}{(2\alpha + 1)(2\alpha + 3)^2} = 1.894240894138507940674283597218 \dots.\end{aligned}\tag{8.1}$$

There exists $\eta \in (0, 1)$ such that

$$\mathbb{E}L_n = \mu n + \delta + O(\eta^n), \quad \text{Var}(L_n) = \sigma^2 n + O(1).\tag{8.2}$$

Moreover,

$$\frac{L_n - \mu n}{\sqrt{\sigma^2 n}} \xrightarrow{d} \mathcal{N}(0, 1).\tag{8.3}$$

8.1 Dominant-root expansion with a size variable

For y near 1, let $\alpha(y)$ be the analytic root near α of

$$\alpha(y)^3 = y\alpha(y) + y^2.\tag{8.4}$$

The root is simple at $y = 1$, so the implicit-function theorem applies. The other two roots have strictly smaller moduli at $y = 1$ and retain a uniform modulus separation in a sufficiently small complex neighborhood of 1. Put $\rho(y) = 1/\alpha(y)$ and $t(y) = y\rho(y)$.

By the same partial fraction calculation as in (7.7),

$$f_n(y) = 2ny^2 \frac{2 + t(y)}{2 + 3t(y)} \alpha(y)^n (1 + O(\eta^n)),\tag{8.5}$$

uniformly in a sufficiently small complex neighborhood, after choosing some $\eta < 1$. The coefficient is nonzero there. This calculation can also be read directly from (4.11); the low-degree correction has no effect on $n \geq 3$.

Set $y = e^s$, and define

$$\lambda(s) = \log \alpha(e^s), \quad g(s) = 2e^{2s} \frac{2 + t(e^s)}{2 + 3t(e^s)}.\tag{8.6}$$

Then the moment generating function satisfies

$$\mathbb{E}e^{sL_n} = \frac{g(s)}{g(0)} \exp\{n(\lambda(s) - \lambda(0))\} (1 + O(\eta^n)).\tag{8.7}$$

All functions and the error are analytic on a fixed small disk. Consequently, after shrinking the disk if needed, derivatives of the logarithmic error are also exponentially small by Cauchy's estimates.

8.2 Computing the cumulants and the normal limit

Differentiate (8.4), remembering $y = e^s$ and $t = y/\alpha(y)$. The resulting identities are

$$\lambda'(s) = \frac{1+2t}{2+3t}, \quad \frac{dt}{ds} = \frac{t(1+t)}{2+3t}, \quad \lambda''(s) = \frac{t(1+t)}{(2+3t)^3}. \quad (8.8)$$

At $s = 0$, $t = \alpha^{-1}$. Using $\alpha^3 = \alpha + 1$ gives $\lambda'(0) = \mu$ and $\lambda''(0) = \sigma^2 > 0$. Also

$$(\log g)'(0) = 2 - \frac{4\alpha^4}{(2\alpha+1)(2\alpha+3)^2} = \delta.$$

Taking the first two logarithmic derivatives of (8.7) proves (8.2).

For completeness, fix a real u and substitute $s = iu/\sqrt{\sigma^2 n}$ in (8.7). Taylor expansion gives

$$n(\lambda(s) - \lambda(0) - \mu s) = -\frac{u^2}{2} + O(n^{-1/2}), \quad \frac{g(s)}{g(0)} \rightarrow 1.$$

Thus the characteristic function of $(L_n - \mu n)/\sqrt{\sigma^2 n}$ converges to $e^{-u^2/2}$, the characteristic function of a standard normal variable. The characteristic-function continuity theorem proves (8.3). This completes the proof of theorem 8.1.

n	$\mathbb{E}L_n$	$\mathbb{E}L_n - \mu n$	$\text{Var}(L_n)/n$
10	7.818181818	1.933137705	0.014876033
20	13.664804469	1.894716243	0.016728567
50	31.319461450	1.894240883	0.016997732
100	60.744682028	1.894240894	0.017038681
200	119.595123162	1.894240894	0.017059155
500	296.146446563	1.894240894	0.017071439
1000	590.398652231	1.894240894	0.017075534

Table 3: Moments computed from the exact binomial distribution, with decimals rounded only for display. The last two columns approach δ and σ^2 .

8.3 Further consequences of the same expansion

Let J_n indicate that $T_n[D]$ is unicyclic. Retaining v in (4.11) replaces the factor $(2+t)/(2+3t)$ in (8.5) by $(2+vt)/(2+3t)$. It does not change the dominant root. Consequently the centered, scaled size and the topology indicator are asymptotically independent:

$$\left(\frac{L_n - \mu n}{\sqrt{\sigma^2 n}}, J_n \right) \xrightarrow{d} (Z, J), \quad (8.9)$$

where Z is standard normal, J is Bernoulli with parameter $1/(2\alpha+1)$, and Z, J are independent. To see the independence directly, put $y = e^{iu/\sqrt{\sigma^2 n}}$ and $v = e^{iw}$. After centering the size, the joint characteristic function tends to

$$e^{-u^2/2} \frac{2 + e^{iw}/\alpha}{2 + 1/\alpha},$$

the product of the two limiting characteristic functions.

Higher fixed-order cumulants require no new enumeration. With

$$\mathcal{L} = \frac{t(1+t)}{2+3t} \frac{d}{dt}, \quad t_0 = 1/\alpha,$$

the r th cumulant is, for every fixed $r \geq 1$,

$$\kappa_r(L_n) = n \mathcal{L}^{r-1} \frac{1+2t}{2+3t} \Big|_{t=t_0} + \mathcal{L}^r \log \frac{2+t}{2+3t} \Big|_{t=t_0} + 21\{r=1\} + O(\eta_r^n) \quad (8.10)$$

for some $\eta_r < 1$. This is an explicit algorithm for all fixed cumulant asymptotics; the first two yield the mean and variance above.

9 Algorithms and reproducible verification

9.1 Exact counting

The main recurrence gives a very small implementation. The following function computes a_n using only integer arithmetic.

```
def trapezohedral_count(n: int) -> int:
    if n < 3:
        raise ValueError("n must be at least 3")
    p = [0] * (n + 1)
    p[0] = 1
    for m in range(1, n + 1):
        p[m] = (p[m-2] if m >= 2 else 0)
        p[m] += (p[m-3] if m >= 3 else 0)
    return 2*n*(2*p[n-2] + p[n-3])
```

The full implementation in `code/research.py` additionally supplies the size/type distribution, graph construction, an independent graph-based membership test, structural enumeration, exact sampling, and data export. It uses only the Python standard library.

Computing all terms through $n = N$ requires $O(N)$ integer additions; the integers have $O(N)$ bits. Thus this is not a claim of linear bit complexity: with ordinary addition the total bit cost is $O(N^2)$. A rolling recurrence needs only a constant number of such integers. A single distant term can instead be obtained by binary powering of the recurrence matrix

$$M = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} b_{m+1} \\ b_m \\ b_{m-1} \end{pmatrix} = M \begin{pmatrix} b_m \\ b_{m-1} \\ b_{m-2} \end{pmatrix},$$

using $O(\log n)$ big-integer matrix multiplications. The integers still have $\Theta(n)$ output bits; matrix powering does not make the bit complexity logarithmic in n .

9.2 An exactly uniform sampler

For a fixed n , choose k with probabilities $c_{n,k}/a_n$. Given k , choose one of the two poles fairly, choose a cyclic zero set Y uniformly from the family counted by (4.3), and choose one of its $4k - n$ eligible

edges uniformly. Reconstruct D by (2.2), using the pole-exchange automorphism when the selected pole is S .

To choose Y uniformly, choose a starting index uniformly from $\{0, \dots, n-1\}$ and choose a uniformly random ordering of s copies of 2 and t copies of 3. The partial sums of these gaps, starting at that index, give the zeros. Every zero set has exactly k representations, so the resulting distribution is uniform even for periodic words. A uniformly shuffled list of repeated gaps is adequate: every distinct gap word has the same number $s!t!$ of preimages under permutations of labeled copies.

For a particular D of size $n - k + 2$, the product of its selection probabilities is

$$\frac{c_{n,k}}{a_n} \cdot \frac{1}{2} \cdot \frac{k}{n \binom{k}{t}} \cdot \frac{1}{4k-n} = \frac{1}{a_n}.$$

This proves exact uniformity. The code draws integers against exact cumulative weights and never converts the weights to floating point. After the size weights have been computed, gap construction uses $O(n)$ storage and $O(n)$ elementary list operations. Validity tests of sampled outputs are useful code checks, but the proof above, not a frequency experiment, establishes uniformity.

9.3 An independent exhaustive check

The C++17 program `code/exhaustive.cpp` constructs the graph directly from (1.1) and examines *every* vertex subset for each requested n . It does not use the classification to prune candidates. For each subset D it checks domination and the private-neighbor criterion by closed-neighborhood bit masks, then checks induced connectivity by a graph traversal. It outputs the complete accepted vertex masks, their sizes, and their induced edge counts.

The Python verifier independently builds every set predicted by [theorem 2.4](#) and compares the resulting sets of bit masks exactly. It also compares the full size/topology distribution against (4.8). The recorded run gives: There is also a separate Python-only exhaustive check for $3 \leq n \leq 7$ inside

n	$ V(T_n) $	All subsets examined	Accepted sets
3	8	256	6
4	10	1,024	16
5	12	4,096	30
6	14	16,384	36
7	16	65,536	70
8	18	262,144	96
9	20	1,048,576	144
10	22	4,194,304	220
11	24	16,777,216	308
Total		22,369,536	926

Table 4: Exhaustion of all subsets, with no reliance on the marked-cycle classification in the acceptance predicate.

`code/research.py`. The much faster exact formulas are cross-checked through $n = 1000$, including the size recurrence, total and topology counts, endpoint multiplicities, divisibility, and strict

log-concavity. All 40 terms displayed in the retrieved OEIS entry agree with the formulas. Finally, 900 sampled sets at nine sizes from $n = 3$ to 100 pass the independent graph-based membership test.

9.4 A third, independent standard-library exhaustion

The archive carries a second, separately written exhaustive verifier, `code/verify.py`, which uses only the Python standard library and shares no code with either `code/research.py` or `code/exhaustive.cpp`. Its independence is the point of keeping it: it builds the graph from (1.1) under the one-based labels $u, v, a_1, \dots, a_n, b_1, \dots, b_n$, and applies the literal definition — coverage of every vertex, failure of every one-vertex deletion to dominate, and induced connectivity — with no appeal to [theorem 2.1](#) or to [theorem 2.4](#) as a filter.

Its coverage test is an incremental dynamic program over bit masks. If d is one set bit of a mask D , then

$$\text{coverage}(D) = \text{coverage}(D \setminus \{d\}) \text{ OR } N[d], \quad (9.1)$$

so processing masks in increasing numerical order makes every smaller coverage mask already available. Connectivity is a separate traversal restricted to the selected vertices. A second routine generates the sets predicted by the cyclic word and mark description, and the program compares the resulting *families of subsets*, not merely their cardinalities. The same program expands the conjectured

n	$ V(T_n) $	Subsets examined	Count a_n
3	8	256	6
4	10	1,024	16
5	12	4,096	30
6	14	16,384	36
7	16	65,536	70
8	18	262,144	96
9	20	1,048,576	144
10	22	4,194,304	220

Table 5: The archived run of the standard-library verifier `code/verify.py`. Exhaustive checking ends at $n = 10$ in this run; the C++ program of [section 9](#) extends the same check to $n = 11$ in a different language. Every value shared by the two runs agrees, at the level of the individual accepted subsets.

rational function by exact formal division, evaluates (1.4) and (4.6), checks both recurrences with their correct starting indices, and checks that summing (4.5) over sizes gives the total; these agree for every $3 \leq n \leq 1000$. All 40 published terms, covering $3 \leq n \leq 42$, agree. It also checks (7.11) and $\alpha > 64/49$ in exact integer and rational arithmetic, and then verifies (7.9) numerically for $45 \leq n \leq 1000$ at high Decimal precision. Those decimal asymptotic calculations are diagnostics, not interval certificates; the nearest-integer claim has a separate uniform proof, [theorem 7.3](#). Its archived private-neighbor certificates for all thirty sets at $n = 5$ are the file cited in [theorem 2.5](#). Computing the first N terms from the recurrence takes $O(N)$ integer arithmetic operations; these are arithmetic-operation counts, not unit-cost bit-complexity claims for unbounded integers.

9.5 What the computation does and does not establish

The exhaustive checks are strong guards against a mistaken graph model, a reversed minimality convention, a missing small case, an incorrect symmetry factor, or an erroneous size refinement. They do not prove a statement about infinitely many graphs. That role belongs to [theorem 2.4](#) and the marked-gap bijection of [section 3](#). Likewise, agreement with all current OEIS terms is a useful external cross-check, not the logical basis of the generating function. The generating function is proved by the mathematical argument, not by finite agreement alone.

9.6 Symbolic and numerical audit

The optional script `code/verify_symbolic.py` uses SymPy and mpmath. It checks equality with the conjectured rational function, the trivariate coefficients through $n = 15$, the weighted Perrin identity, reduction of the asymptotic constants modulo $\alpha^3 - \alpha - 1$, and the exact rational certificate in the rounding proof. A multiplication matrix for the cubic number field provides an independent exact trace check of the Binet weights. At 180-decimal working precision it also checks the Binet and nearest-integer formulas through $n = 1000$. These numerical tests are supplementary: the algebraic partial fractions and rational inequalities establish the formulas for all indices.

The saved environment used Python 3.13.5, g++ 14.2.0, SymPy 1.14.0, mpmath 1.3.0, and pdfTeX from TeX Live 2025/dev. The standard-library code is written for Python 3.9 or later. No network access or paid service is needed to rerun it.

9.7 Reproduction commands and archive contents

From the archive's top-level directory, run:

```
g++ -O3 -std=c++17 code/exhaustive.cpp -o exhaustive
./exhaustive 11 > data/exhaustive.csv
python code/research.py verify --exhaustive-csv data/exhaustive.csv
python code/research.py data --maximum 1000
python code/verify.py --brute-max 10 --max-n 1000
python code/verify_symbolic.py
latexmk -pdf -interaction=nonstopmode article.tex
```

The third command is the independent standard-library verifier of [section 9.4](#); run it with ordinary Python, not `python -O`, because it implements its checks as assertions. External Python packages are needed only for the optional symbolic audit; they are listed in `requirements-optional.txt`. The archive includes the complete article source and PDF, three independent implementations of the acceptance test, a 998-row term table covering $3 \leq n \leq 1000$, two independently produced size distributions through $n = 100$, all exhaustive accepted masks, private-neighbor certificates at $n = 5$, and machine-readable verification reports. A ready-to-paste OEIS formula/proof note is supplied separately. No claim has been submitted to OEIS automatically.

10 What has been established

The conjectured rational function (1.2) follows from an explicit bijection valid for every $n \geq 3$. The central mechanism is not extrapolation from computed terms: private neighbors force one selected

pole and one opposite-class vertex, then convert domination and minimality into local constraints on a cyclic word. Those constraints are exactly length-2 and length-3 zero gaps with a restricted edge marker.

Once this classification is proved, the rest of the theory is determined. Cutting at the marked gap gives the Padovan-type formula, fixing gap counts gives the exact size distribution, and distinguishing the marked gap type gives the induced topology. The size distribution is strictly log-concave; the total count has a complete three-root expression with an exponentially decaying absolute correction; the random size has explicit linear mean and variance and a Gaussian limit.

The finite graph checks, exact symbolic calculations, and independent implementations support the derivation, but no finite computation is used as a substitute for its all- n proof. The main remaining external question is bibliographic priority: the public record inspected here still calls the formula conjectural, while a targeted literature search is not an exhaustive search of unpublished work or private communications.

A A compact proof for the sequence record

Here is the mathematical core without the refinements.

In T_n , let N, S be the poles and let the alternating ring be $a_0b_0 \cdots a_{n-1}b_{n-1}a_0$. A connected minimal dominating set cannot contain both poles, since they already dominate and are disconnected. It cannot omit both: a connected ring-only dominating set contains a path of at least $2n - 2 \geq 4$ vertices, whose internal vertices have no private neighbors. Assume it contains N . It must contain some b to dominate S ; each selected b needs a selected a for connectivity and can have only S as a private neighbor. Thus exactly one b_j is selected.

Write $x_i = 1$ for selected a_i . Domination and connectivity exclude 00. A selected a_i needs an unselected adjacent $b \neq b_j$ with its other a -neighbor unselected; thus 111 is excluded and the marked edge (a_j, a_{j+1}) must be either the middle 11 of 0110, or one edge of 010. Conversely these conditions provide private neighbors for every selected vertex, including S for b_j and a zero not incident to the marker for N .

Zeros therefore have cyclic gaps 2 or 3. A marked length-2 gap has two orientations and a marked length-3 gap one. Fixing the pole and b_j , then removing the marked gap, gives $2p_{n-2} + p_{n-3}$ compositions into 2 and 3. There are $2n$ pole/marker choices. Hence $a_n = 2n(2p_{n-2} + p_{n-3})$, and differentiating $(2x^2 + x^3)/(1 - x^2 - x^3)$ and deleting the artificial $n = 2$ term gives the stated generating function.

B Source audit and attribution

The sources were consulted on 19 September 2026. The main entry's internal record was revision #16, timestamped 7 January 2026 at 10:23:48. It attributes the sequence to Eric W. Weisstein, the generating-function conjecture to Joerg Arndt, and the extension beginning at $a(14)$ to Christian Sievers [1]. Its general site footer is not evidence of a later revision to the mathematical claim itself.

Searches included the exact sequence identifier, the phrase "connected minimal dominating" with "trapezohedral", and combinations of the graph family with "Arndt", "proof", and "generating function". They located the conjectural record but no earlier proof. This is the basis for selecting the

problem as apparently unresolved in the available public record. The known Padovan composition interpretation and Perrin maximal-independent-cycle interpretation are explicitly credited rather than claimed as new. The proof and refinements in this manuscript were developed in the present investigation.

C Provenance of this archive

This package is the union of two independently prepared manuscripts on A381190 that turned out to contain the same theorem established by the same argument. The text above is this package's original manuscript; a second, incoming manuscript has been folded into it. The classification of section 2 and the marked-gap bijection of section 3 are kept as the *single* proof. The incoming manuscript's "marked-block" presentation is that same bijection read one block at a time rather than one gap at a time, and is recorded as such in section 3; it is deliberately not offered as an alternative proof, because there is only one argument here.

Carried across from the incoming manuscript, and restated above in the notation used here rather than in its own, are: the finite binomial sum theorem 4.2; the unnormalized nearest-integer formula theorem 7.3 with its integer certificate (7.11); the expanded size-refined rational form theorem 4.5; the explicit evaluations $Q(0) = 1$, $Q(-2/3) = 23/27$ and $Q(-2) = 5$ supporting the minimality of order 6; the worked certificate of theorem 2.5; the error table table 2; and the third independent verifier of section 9.4. Three notational conversions had to be applied, and each one would silently corrupt an imported formula if omitted.

- The incoming manuscript wrote (r, s) for the numbers of length-2 and length-3 blocks. Those are the quantities written (s, t) here, as in (4.1); in particular its s is this text's t , not this text's s . Every imported formula was rewritten accordingly, and its "number of legal marks" $2r + s$ is the quantity $2s + t = 4k - n$ of (4.4).
- It wrote C for the leading asymptotic constant and λ for the plastic constant. Its C is the constant written $K = 2C(\alpha)$ here, which is *twice* the $C(\alpha)$ of (7.3); its λ is α .
- It indexed the size-refined counts by the cardinality d , writing $a_{n,d}$ where this text writes $c_{n,k}$ with $k = n - d + 2$. Under that substitution $3n - 4d + 8 = 4k - n$ and $2d - n - 4 = n - 2k$, so the two formulas and the two size ranges agree identically; its sample values $a_{12,8} = 48$, $a_{12,9} = 384$, $a_{12,10} = 24$ are $c_{12,6}$, $c_{12,5}$, $c_{12,4}$ in (4.5).

The two packages' term tables agree digit for digit for all $3 \leq n \leq 1000$, and their exhaustive runs agree at the level of individual accepted subsets for every n they share, so no reconciliation of data was required.

The incoming manuscript also recorded how its author came to this problem: a pre-registered four-entry shortlist and one operating-system random draw that returned its fourth entry, with no rerolls. That record is provenance for that manuscript's problem selection, not a statement about A381190, so it is retained outside this text in the archive's provenance/ directory and is not restated here. Its randomness came from the operating system and was not generated from a published deterministic seed. The record therefore documents the actual selection, not a seed-replayable experiment. The three unselected conjectures on that shortlist are not claimed to be solved here.

References

- [1] OEIS Foundation Inc., *A381190: Number of connected minimal dominating sets in the n -trapezohedral graph*. Sequence by Eric W. Weisstein, 16 February 2025; generating-function conjecture by Joerg Arndt, 7 January 2026. <https://oeis.org/A381190>. Internal record: <https://oeis.org/A381190/internal>.
- [2] Eric W. Weisstein, *Trapezohedral Graph*, MathWorld—A Wolfram Resource. <https://mathworld.wolfram.com/TrapezohedralGraph.html>.
- [3] Eric W. Weisstein, *Minimal Dominating Set*, MathWorld—A Wolfram Resource. <https://mathworld.wolfram.com/MinimalDominatingSet.html>.
- [4] Eric W. Weisstein, *Minimal Connected Dominating Set*, MathWorld—A Wolfram Resource. <https://mathworld.wolfram.com/MinimalConnectedDominatingSet.html>.
- [5] OEIS Foundation Inc., *A000931: Padovan sequence*. See in particular the shifted composition interpretation with parts 2 and 3. <https://oeis.org/A000931>.
- [6] OEIS Foundation Inc., *A001608: Perrin sequence*. See the interpretation as maximal independent sets in a cycle, for $n \geq 3$. <https://oeis.org/A001608>.